

In the **SARTA** (Stand-alone AIRS Radiative Transfer Algorithm) model, the "predictor set" refers to the specific atmospheric state variables used in its linear regression parameterization to calculate the optical depth of absorbing gases for satellite sensor channels. [1, 2, 3, 4]

SARTA relies on a polynomial and statistical regression approach to rapidly compute transmittances rather than relying entirely on slow, line-by-line integration. The exact predictor set typically used for its gas absorption algorithms consists of the following variables: [3, 5, 6]

- **Layer-mean kinetic temperature (T):** Used for computing the temperature dependence of absorption lines.
- **Absorber amount (u):** The column amount of the target gas (such as H₂O, O₃, CO₂, CH₄, N₂O, CO, HNO₃, etc.) for a given atmospheric layer.
- **Pressure (P) and Layer Boundaries:** Used to account for the pressure broadening of spectral lines.
- **Secant of the satellite viewing angle:** Used to account for the path length of the radiation traveling through the atmosphere.

How SARTA Predictors Work

1. **Regression Coefficients:** SARTA uses pre-calculated regression coefficients derived from accurate line-by-line models (such as LBLRTM) and spectroscopic databases (e.g., HITRAN). [3, 5, 7, 8]
2. **Predictor Polynomials:** The predictor variables are evaluated using a series of mathematical terms (such as differences from a reference profile, squared terms for temperature variations, and absorber amount combinations). [3]
3. **Fast Computation:** By evaluating the predictor equations, SARTA rapidly generates layer-to-space transmittances for thousands of instrument channels without computing every single spectral line in real-time. [5, 9]

Would you like to know more about how SARTA handles cloud or aerosol scattering in its radiative transfer calculations?

[1] <https://journals.ametsoc.org>

[2] <https://gmd.copernicus.org>

[3] <https://www.researchgate.net>

[4] <https://repository.library.noaa.gov>

[5] <https://www.sciencedirect.com>

[6] <https://www.sciencedirect.com>

[7] <https://www.sciencedirect.com>

[8] <https://ntrs.nasa.gov>

[9] <https://www.ecmwf.int>